transformers package.
- *Use case:* Developing a lightweight language model for tasks like domain-specific NER or text categorisation.
- *Advantages:*
  - Lightweight and efficient for production use.
  - Easy to train and deploy custom pipelines.

AllenNLP
- *Overview:* AllenNLP is a research-focused NLP library built on PyTorch, designed for experimenting with and building custom NLP models.
- *Key features:*
  - Modular design for building and training custom models.
  - Pre-built components for common NLP tasks (e.g., text classification, sequence tagging).
  - Integration with transformer-based models.
- *Use case:* Prototyping and training custom NLP models for academic research or specialised tasks.
- *Advantages:*
  - Research-oriented with support for custom architectures.
  - Strong focus on explainability and visualisation.

Fairseq
- *Overview:* Fairseq is a sequence-to-sequence modelling toolkit by Facebook AI Research, supporting tasks like machine translation and text generation.
- *Key features:*
  - Support for transformer and other sequence-to-sequence architectures.
  - Customisable training pipelines.
  - Pre-trained models for translation, summarisation, and more.
- *Use case:* Developing a specialised translation model for low-resource languages or domain-specific text generation.
- *Advantages:*
  - High flexibility for sequence-to-sequence tasks.
  - Scalable for large datasets and distributed training.

OpenNMT
- *Overview:* OpenNMT is an open source toolkit for training neural machine translation models, but it can also be adapted for other NLP tasks.
- *Key features:*
  - Support for training custom models from scratch.
  - Tools for preprocessing and tokenizing text data.
  - Flexible architecture for incorporating domain-specific data.
- *Use case:* Training a translation model for legal or technical documents.
- *Advantages:*
  - Highly customisable for domain-specific tasks.
  - Active support for multilingual applications.

T5 (Text-to-Text Transfer Transformer)
- *Overview:* T5 is a transformer model by Google that treats all NLP tasks as text-to-text problems.
- *Key features:*
  - Unified framework for tasks like classification, translation, and summarization.
  - Pre-trained models available for fine-tuning.
  - Open source implementation via Hugging Face and TensorFlow.
- *Use case:* Training a model for domain-specific summarization or question answering.
- *Advantages:*
  - Versatile framework for multiple NLP tasks.
  - Easy to fine-tune for specialised use cases.

FastText
- *Overview:* FastText is a lightweight library by Facebook for text classification and word representation.
- *Key features:*
  - Efficient training of word embeddings.
  - Support for supervised and unsupervised learning.
  - Works well with small datasets.
- *Use case:* Building a domain-specific text classifier or embedding model for real-time applications.
- *Advantages:*
  - Extremely fast and efficient.
  - Suitable for resource-constrained environments.

Cohere (open source support)
- *Overview:* Cohere offers APIs for language model training and inference, with open source support for model customization.
- *Key features:*
  - Pre-trained models for text generation and classification.
  - Fine-tuning for domain-specific applications.
  - Integration with open source NLP tools.
- *Use case:* Creating a specialised language model for marketing or creative content generation.
- *Advantages:*
  - Easy-to-use APIs for rapid experimentation.
  - Focus on practical applications.

SentenceTransformers
- *Overview:* A library for building and fine-tuning sentence embeddings using transformer models.
- *Key features:*
  - Pre-trained models for generating semantic embeddings.